

Modeling the Effect of Pitch Sequencing and Batter-Specific Adjustments on Swing-and-Miss Outcomes in MLB

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1 Introduction

The rise of advanced tracking systems in all sports, especially Major League Baseball (MLB) has transformed how teams evaluate pitching performance and decide on their pitching strategy. Since the introduction of Statcast in 2015, every pitch thrown in any MLB game has been measured through many different metrics such as release speed, spin rate, movement profile, release point, and extension. All these metrics are captured at the millisecond (Major League Baseball, 2024). This sudden increase in analytics has led to massive changes in pitcher development, with front offices investing heavily in how pitchers are drafted and managed throughout their tenure. All of this is done to maximize the effectiveness of each pitch in a pitcher’s arsenal. This leads to the central question in baseball analytics: *what attributes make a pitch difficult to hit?* Intuitively, batters and most baseball watchers agree that pitches with higher velocity, greater movement, and longer extension are harder to hit. However, standard logistic regression models that predict swing-and-miss probability based solely on these physical characteristics often lead to inconclusive results. In such models, the probability of a swing-and-miss, $p_i = P(Y_i = 1)$, is typically described as:

$$\log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 \text{Speed}_i + \beta_2 \text{IVB}_i + \beta_3 \text{HB}_i + \beta_4 \text{Ext}_i + \beta_5 \text{Arm}_i + \beta_5 \text{Arm}_i \quad (1)$$

where IVB denotes induced vertical break, HB horizontal break, Ext release extension, and Arm arm angle. Previous such studies have also found that main effects and simple interactions in this framework are often statistically insignificant or fail to explain the variance in outcomes. The limitation of this approach is that it treats each pitch as an independent, context-free event, ignoring two critical sources of variation: **sequencing** and **batter heterogeneity**. First, pitches are part of a strategic sequence. A pitch’s effectiveness cannot be explained solely based on its individual metrics, but rather the buildup to that pitch. For example, a breaking ball may induce more whiffs when it follows a fastball due to the disruption in batter timing and visual processing, a phenomenon also known as pitch tunneling. Second, batters differ significantly in their baseline contact rates. Elite contact hitters like Shohei Ohtani or Aaron Judge rarely swing and miss regardless of pitch

quality, while high-strikeout batters may whiff at average pitches. Neglecting these batter-specific tendencies introduces a lot of noise into the analysis of this dataset, which in turn hinders the conclusions that we can draw about the true relationship between pitch quality and outcomes. To address these limitations, we want to propose a refined framework that tackles this issue by incorporating pitch sequencing and batter heterogeneity. Specifically, our research question is:

How do pitch sequencing (the type, location, and result of the previous pitch) and batter-specific tendencies jointly influence the probability of a swing-and-miss, beyond the physical characteristics of the pitch itself?

To investigate this, we propose the following objectives:

1. **Pitch Sequence Effects:** Investigate whether the previous pitch type and result significantly affect the whiff probability of the current pitch. We believe that specific sequence combinations (e.g., Fastball $\rightarrow$ Slider) will show significant interaction effects.
2. **Differentiate between batter skill and Pitch quality:** Implement a Generalized Linear Mixed Model (GLMM) where individual batters are treated as random effects. This will help separate the variance attributable to the pitch from the variance attributable to the batter’s skill.
3. **Prediction Power:** Evaluate whether the Sequencing and Mixed-Effects models provide a better fit and have more predictive power compared to the baseline metrics that a single pitch provides.

2 Research Strategy

2.1 Data

2.2 Planned Analysis

References

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